Nathan Xia

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Engineering Portfolio: Nathan Xia

Education

- **University of Toronto** — Toronto, Canada
  Bachelor of Applied Science - Engineering Science — *September 2025 - Present*

  cGPA: 3.95/4.00
  Relevant Courses: Vector Calculus and Fluid Mechanics, Thermodynamics and Heat Transfer, Ordinary Differential Equations, Linear Algebra

Projects

- **Lattice Boltzmann CFD Solver**: This project is ongoing. View Results.
  - Developed and verified a Python-based D2Q9 Lattice Boltzmann CFD solver from scratch featuring configurable geometries and flow conditions, velocity, pressure, and vorticity visualization, and aerodynamic force tracking.
  - Benchmarked cylinder-flow behaviour at $Re = 150$ yeilding a C_d **within 1.4% of a published result**, **reproducing von Kármán vortex shedding**, and the expected oscillations in lift and drag forces.
  - Further benchmarked the solver on a NACA 0012 at $Re = 2000$ and $AOA = 8^{\circ}$, **producing a Strouhal number within 1.8% of a published ANSYS Fluent result.**
- **Vortex Generator Research Project: Paper**
  - Designed and conducted a vortex-generator study on a NACA 2412 airfoil using a self-designed and manufactured flow-visualization wind tunnel.
  - Identified an optimal gap width and chordwise placement; regression predicted a **76% increase in the angle at which separation reached 30% chord**.
  - Designed the airfoil in Desmos and CAD and presented the findings at CYSF and CWSF.
  - Selected as **1 of 15 projects from** ≈ 600 to compete at CWSF and earned a bronze medal among ≈ 350 projects.
- **ANSYS Fluent Analysis — Vortex Generator Project Extension**: View Details.
  - Extended the study with **3D steady laminar ANSYS Fluent simulations** at $Re \approx 4{,}300$, reproducing the wind-tunnel geometry and no-slip boundaries.
  - Built locally refined polyhedral meshes with boundary-layer inflation and verified convergence using residuals, force histories, and mass balance.
  - A **two-mesh sensitivity study** produced differences of **0.9% in lift coefficient, 1.4% in drag coefficient, and 1.3 mm in separation location**.
  - Compared control and VG separation using signed wall shear stress and identified force-coefficient sensitivity to domain confinement.

Experience

- **U of T Institute for Aerospace Studies** — FCET Research Group
  Research Intern — *May 2026 - September 2026*
  - Selected as **1 of only 2 first-year students** to conduct research at UTIAS, independently leading an aeroacoustics study on remote microphone probe calibration for wind-tunnel pressure measurements. Found here: Preprint
  - Owned the full experimental workflow from apparatus and test-matrix design through **data acquisition and FFT-based signal processing** across 16 investigated variables, producing a **70+ GB dataset across 1,200+ collected data files.**
  - Developed a **MATLAB** acquisition and analysis pipeline and identified improvements that **reduced transfer function magnitude standard deviation by 69%** and **extended the usable frequency range to 1.5**$\times$ its original range.
  - Experimentally validated findings through **20+ trials** that informed the group's ongoing RMP research and future RMP projects, including improved calibration configurations and procedures.
- **Robotics for Space Exploration - Aerial** — University of Toronto
  Mechanical Co-Lead — *July 2026 - Present*
  - Direct all mechanical design and integration activities for the subsystem, with responsibility for architecture, **CAD**, analysis, manufacturing, assembly, and final system integration.
  - Lead a team of mechanical recruits, overseeing **CAD onboarding, CFD analysis,** design iteration, and manufacturing while reviewing and coordinating technical work across the team.

  Mechanical Sub-team Recruit — *September 2025 - July 2026*
  - Conceived the glider's lifting-body configuration, replacing a conventional fuselage with an airfoil-profile body; evaluated **20+ airfoils in OpenVSP/VSPAERO** to select the final geometry.
  - Integrated vortex generators and end plates into the glider design, achieving **5**$\times$ **the simulated lift at 20 m/s relative to the baseline configuration without these devices.**
  - Designed and simulated rocket nose-cone geometries in Autodesk CFD, iterating through **30+ profiles** using CFD and rocket-performance simulations.

Skills Summary

- **Tools:** ANSYS Fluent, ANSYS Mechanical (FEA), Autodesk CFD, SolidWorks, Fusion 360, Autodesk Inventor, OpenRocket, OpenVSP
- **Skills:** Wind tunnel analysis and design, CFD + FEA simulation and meshing, aerodynamic analysis and design, CAD/CAM, signal processing and analysis
- **Languages:** Python, MATLAB, C, Java, C++ (Arduino), System Verilog